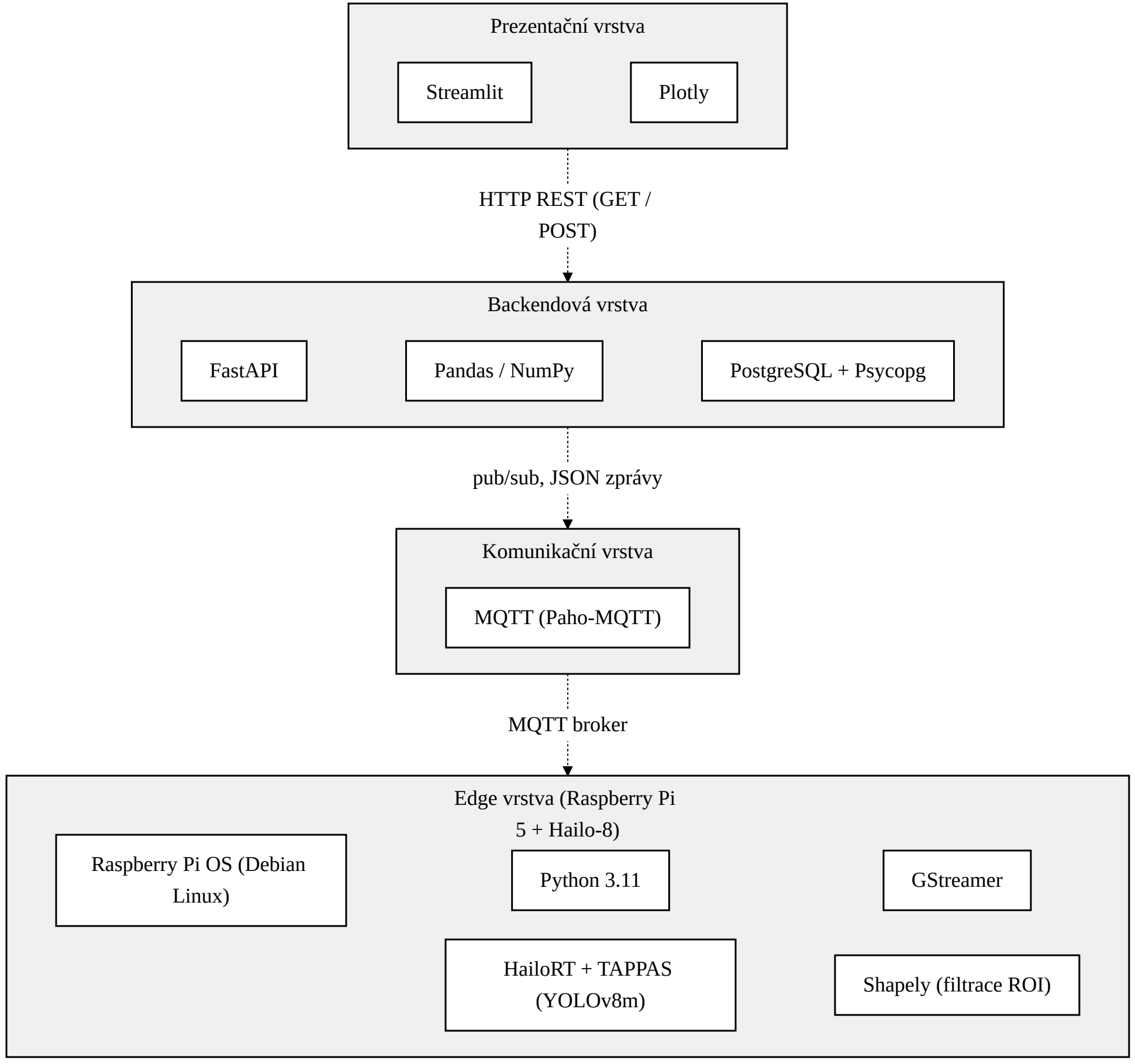
Prezentační vrstva
Streamlit
Plotly
HTTP REST (GET / POST)
Backendová vrstva
FastAPI
Pandas / NumPy
PostgreSQL + Psycopg
pub/sub, JSON zprávy
Komunikační vrstva
MQTT (Paho-MQTT)
MQTT broker
Edge vrstva (Raspberry Pi 5 + Hailo-8)
Raspberry Pi OS (Debian Linux)
Python 3.11
GStreamer
HailoRT + TAPPAS (YOLOv8m)
Shapely (filtrace ROI)